

lists. Clothing embedded with RFID tags has been used to operate doors that detect the tag and open in response to it. Space suits used by astronauts are dense with miniaturised electronics used for a variety of purposes.

Therefore clothing integrated with electronics exists today, but it cannot really be called 'smart.' Users, in general, seek smart clothing that possesses many of the useful properties of computers—the ability to store and manipulate data, display images, text and video, connect to the Internet, offer input devices and so on. Other possible features of smart clothing include the ability to detect chemicals in the air, quickly harden on contact with a speeding bullet, change colour or opacity, generate power from the wearer's movement, record the wearer's speech and activity, and even project an image of the scene behind the wearer, creating a crude form of camouflage-based invisibility and releasing chilled water vapour when it senses that the wearer is hot.

Trials of shirts made with smart fabrics are already on their way in Europe. These studies will help scientists understand what needs to be improved and what works. These smart fabrics, often called first-generation fabrics, can monitor vital signs like heart rate and temperature.

However, scientists think that these achievements are modest. Much more can be done with smart clothing. Researchers in Europe have successfully developed techniques that make the sensors almost invisible and reliable. This new development is exciting for many reasons. Clothes are usually meant to be soft, light, thin and flexible, whereas most electronic components are hard, heavy, thick and rigid, which makes it difficult to integrate electronics into clothing. There is a need for advanced, custom-made electronics having properties desirable for integration into clothing and continuous efforts are being made towards developing such advanced electronics. It is commonly thought that flexible displays will be

one of the first big steps toward smart clothing.

Making of electronic textiles

The idea of electronic yarns and textiles has appeared for quite some time, but their properties often do not meet practical expectations. In addition to chemical/mechanical durability and high electrical conductivity, important material qualifications include weavability, wearability, light weight and 'smart' functionalities.

transformed into intelligent e-textiles by coating them with carbon nanotubes (CNTs). Efficient charge transport through the network of nanotubes (20 ohms/cm) and the possibility to engineer tunneling junctions make them promising materials for many high-knowledge-content garments. Along with integrated humidity sensing, it is demonstrated that CNT-based cotton threads can be used to detect albumin—the key protein of blood—with high sensitivity and selectivity.

Carbon nanotubes are cylindrical



Carbon-nanotube coated smart yarn can conduct enough electricity from a battery to power a light-emitting diode device. Researchers can use its conductivity to design garments that detect blood. (Photo courtesy: University of Michigan)

The textile industry can today produce threads that are highly conductive and as flexible as regular fabric so that sensors in garments can measure biometric characteristics like the wearer's temperature or heart rate. Such garments could have important applications in medicine and sports.

However, weaving sensors into these textiles is tricky. Threads not only need to be highly durable to survive the weaving process but, because fabrics are woven on a large scale and then cut to size, it is hard to ensure they will end up where they are needed in the garment.

A team of researchers at the University of Michigan (USA) has come up with a way in which smart clothing can be a reality: Cotton threads can be

carbon molecules with a unique honeycomb-like arrangement of atoms. These are regarded as one of the most versatile nanomaterials available because of their mechanical strength and electrical properties.

Nanotube composites are often made into solid structures or sheets, although flexible versions, such as electrically conductive films and electronic inks, can be prepared from dilute nanotube solutions. Some electronic devices, such as field-emission displays in some flat panels, are made from nanotube yarns. But the weaving of these yarns, which may be only one-thousandth of a millimetre thick, is complicated and expensive. Creating garments with electrical properties is supposed to be the other approach.